
CoPE: Continual Probe-guided Expansion for Large Vision-Language Models

Ziqin Wang^{*1} Hengyuan Zhao^{*1} Qixin Sun^{*1} Kaiyou Song Yilin Li Xiaolin Hu² Qingpei Guo
Linjiang Huang¹ Si Liu^{1,3}

Abstract

Mixture of Experts architectures have recently advanced the scalability and adaptability of Large Language Models for continual multimodal learning. However, extending these models to accommodate sequential tasks remains challenging. As new tasks arrive, naive model expansion leads to rapid parameter growth, while modifying shared routing components often causes catastrophic forgetting, undermining previously learned knowledge. To address these issues, we propose **CoPE**, a continual learning framework for LLMs that requires no replay data of previous tasks and ensures both parameter efficiency and robust knowledge retention. Our approach introduces the **Probe-Guided Knowledge Extension** mechanism, which uses probe experts to dynamically determine when and where new experts should be added, enabling adaptive and minimal parameter expansion tailored to task complexity. To support inference without task labels, we further incorporate a **Probabilistic Task Locator** that dynamically matches inputs to the correct task-specific components. To handle the practical issue that task labels are unknown during inference, we leverage a VAE-based reconstruction strategy to identify the most suitable router by matching input distributions, allowing automatic and accurate expert allocation. This design mitigates routing conflicts and catastrophic forgetting, enabling robust continual learning without explicit task labels. Extensive experiments on the CoIN benchmark, covering eight diverse VQA tasks, demonstrate that **CoPE** delivers strong continual learning performance with a compact model size, significantly reducing forgetting and param-

eter overhead compared to prior methods. These results showcase the effectiveness and scalability of our approach for parameter-efficient continual learning in large language models. Our code will be available at <https://github.com/wz7in/COPE>.

1. Introduction

Multimodal Large Language Models (Caffagni et al., 2024; Wu et al., 2023; Li et al., 2023; Radford et al., 2021; Liu et al., 2023) often employ Parameter-Efficient Fine-Tuning (PEFT) methods (Abou Baker et al., 2024; Han et al., 2024; Hu et al., 2022; Houlsby et al., 2019; Liu et al., 2022; 2021; Edalati et al., 2022; Zhang et al., 2022; Chen et al., 2022) after large-scale pre-training to efficiently adapt to downstream tasks without full retraining, ensuring computational efficiency and competitive performance. These training processes are commonly conceptualized as a multi-task learning paradigm, where all tasks’ data are simultaneously accessible. However, in real-world scenarios, knowledge and tasks are continuously updated in a streaming manner, posing new challenges for the efficient adaptation of MLLMs. This dynamic environment necessitates that fine-tuning methods not only remain parameter-efficient, but also support robust continual learning to enable models to incrementally acquire new knowledge from new tasks.

Catastrophic forgetting (Kirkpatrick et al., 2017), a fundamental issue in CL, also poses a significant challenge for achieving efficient continual learning in MLLMs. To address this issue, some approaches (Cai et al., 2023; Lei et al., 2023; Yang et al., 2023) involve storing data from previous tasks as subsets or distributions, and utilize a data replay scheme during new tasks training to prevent knowledge forgetting. However, as tasks multiply, the storage and computational overhead associated with replay-based methods can become prohibitive. Another line of works (Lao et al., 2023; Zheng et al., 2023; Farajtabar et al., 2020; Zhu et al., 2021; Chiaro et al., 2020; Serrà et al., 2018; Jha et al., 2024; Cai et al., 2022; He et al., 2023; Peng et al., 2021; Yang et al., 2023) focuses on designing novel loss functions or model architectures to alleviate forgetting. Nonetheless, these approaches typically involve updating parameters shared across tasks, which may still lead to performance

¹Institute of Artificial Intelligence, Beihang University
²Tsinghua University ³Hangzhou Innovation Institute, Beihang University. Correspondence to: Linjiang Huang <ljhuang@buaa.edu.cn>, Qingpei Guo <qqp.hust@gmail.com>.

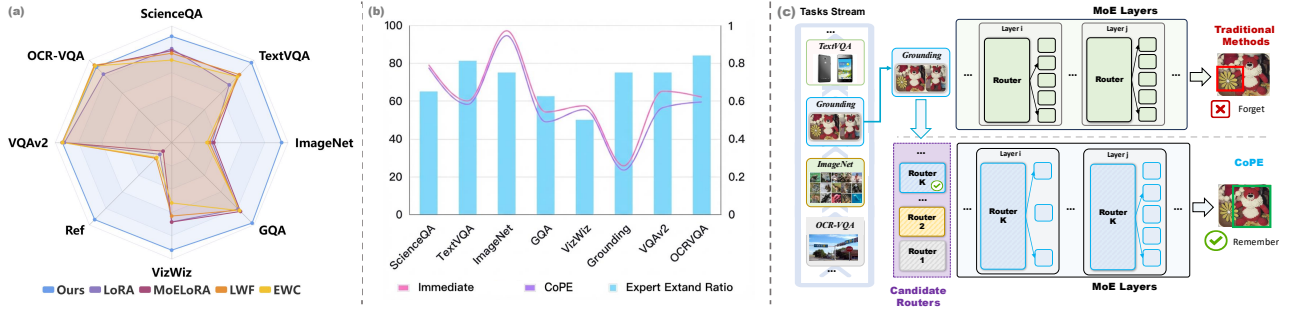


Figure 1. **a) Visualization of the Model’s Anti-Forgetting Capability.** Our method significantly improves anti-forgetting capability, achieving a performance boost compared to baselines. **b) Model Forgetting Ratio vs. Parameter Expansion Rate.** *Immediate* means the performance just after training. After training on task streams (from left to right), our methods nearly match the performance of *Immediate* while saving nearly 30% parameters, enhancing efficiency. **c) Conceptual comparison between the previous method and our CoPE.** Unlike traditional methods that use a fixed number of experts, our model dynamically adjusts experts per layer based on task needs. Besides, we also build a router bank to reduce forgetting and improve knowledge retention. For clarity, we illustrate only the routing and experts within the block.

degradation on previously learned tasks due to interference.

Recently, Wang et al. (Wang & Li, 2024) propose to use a Mixture of Experts architecture, adding new experts at all layers to acquire task-specific knowledge in continual learning. Though MoE’s scalability makes it promising for continual learning, two key challenges remain: 1) **When and where to insert experts?** Existing work (Yang et al., 2024) shows that additional parameter demands vary greatly across tasks, indicating expert allocation should be adaptive to task similarity instead of statically predefined. (Zhong et al., 2024) explore dynamic allocation by analyzing expert selection distribution changes before/after task training, but such shifts only reflect task preference (not real need for new experts), leaving current methods struggling to accurately identify when/where new experts are required. 2) **How to mitigate catastrophic forgetting in the Router?** In MoE, the Router is crucial for dynamic expert allocation. Optimizing experts for new tasks may cause forgetting of prior knowledge, and fine-tuning the Router can alter earlier tasks’ routing strategies, also causing catastrophic forgetting. (Yu et al., 2024) address this by fixing expert count and adding routers continuously to preserve prior routers’ knowledge integrity. However, as tasks increase, a fixed expert set may limit the model’s capacity to accommodate highly different tasks.

Building on the aforementioned challenges, we propose an effective framework, **CoPE**, enabling effective continual learning for MoE-based MLLM without requiring replay data. To address the challenge of adaptive expert expansion, we propose the **Probe-Guided Knowledge Extension (PKGE)** algorithm. This approach utilizes minimal probe data from new tasks to monitor the behavior of newly added expert modules at each layer. By analyzing per-layer probe activation frequencies, the model adaptively decides whether to expand capacity at that layer, preventing unneces-

sary parameter growth (Zhong et al., 2024) while preserving previously acquired knowledge during the learning of new tasks. To mitigate catastrophic forgetting arising specifically from *router updates*, we utilize a reconstruction-based **Probabilistic Task Locator (PTL)** to facilitate automatic router selection. Rather than relying on a single shared router that risks entangling task-specific routing preferences, PTL assigns each task a lightweight, dedicated router that preserves prior routing behavior. Activating a task’s router reinstates its routing policy and task-specific performance with minimal overhead, effectively isolating tasks while maintaining scalability. During inference, when task identities are unknown, we adopt a reconstruction-based strategy inspired by variational methods (An & Cho, 2015; Pu et al., 2016). Using a VAE-based framework, the model learns task-specific distributions and infers task identities by evaluating reconstruction errors, enabling PTL to dynamically route inputs and maintain robust performance on unseen tasks.

We evaluate CoPE on the CoIN dataset (Chen et al., 2024), a benchmark tailored to assess continual learning performance across eight distinct VQA-based tasks. Through extensive quantitative and qualitative evaluations, along with comprehensive ablation studies, we demonstrate that our approach not only significantly mitigates catastrophic forgetting, but also promotes effective knowledge transfer and adaptation as new tasks are learned sequentially.

Conflict of Interest Disclosure. We declare that we have no relevant or material financial interests that relate to the research described in this paper.

2. Related Work

Continual Learning. Continual learning aims to mitigate catastrophic forgetting (Hassabis et al., 2017; Wu

et al., 2024) and enable incremental knowledge acquisition. Existing methods fall into four main categories: 1) *Regularization-based* approaches (Lao et al., 2023; Zheng et al., 2023; Farajtabar et al., 2020; Zhu et al., 2021), which constrain updates to important parameters; 2) *Architecture-based* approaches (Chiaro et al., 2020; Serrà et al., 2018; Jha et al., 2024; Cai et al., 2022; Sun et al., 2021; He et al., 2023; Peng et al., 2021; Yang et al., 2023; Yu et al., 2024; Chen et al., 2023a; Sun et al., 2026), which add task-specific components to reduce interference; 3) *Replay-based* approaches (Cai et al., 2023; Lei et al., 2023; Yang et al., 2023; Lopez-Paz & Ranzato, 2017), which store or generate samples for rehearsal; and 4) *Prompt-based* methods (Wang et al., 2023b; Qian et al., 2023; D’Alessandro et al., 2023; Zheng et al., 2024), which use learnable prompts to maintain performance. However, computational and storage burdens, as well as forgetting due to parameter updates, remain open challenges (Chen et al., 2024).

Mixture of Experts. The MoE architecture (Riquelme et al., 2021; Shen et al., 2023; Mustafa et al., 2022) employs specialized expert networks and a gating mechanism for efficient computation. Sparsely-gated MoE (Shazeer et al., 2017; Lepikhin et al., 2020) has shown strong performance in LLMs, such as Mixtral 8x7B (Jiang et al., 2024), across diverse NLP tasks. Recent work (Liu et al., 2024; Yang et al., 2024; Chen et al., 2023b; Luo et al., 2024) combines MoE with LoRA (Hu et al., 2022) for more efficient training. MoE’s scalability has led to its adoption in continual learning, e.g., LEMoE (Wang & Li, 2024) adds experts at all layers for new tasks, and Lifelong-MoE (Chen et al., 2023a) trains new experts while freezing old ones. CoIN (Chen et al., 2024) further explores MoELoRA, but existing methods still face parameter overhead and suboptimal robustness.

Large Language Models & PEFT. Recently, Large Language Models (Liu et al., 2023; Touvron et al., 2023; Chowdhery et al., 2023; Brown et al., 2020) have garnered widespread attention for their remarkable abilities in areas such as language generation, in-context learning, and reasoning. To enable data- and compute-efficient adaptation for specific downstream tasks, various PEFT methods (Houlsby et al., 2019; Li & Liang, 2021) have been introduced. Among these, LoRA (Hu et al., 2022) stands out by representing weight updates through low-rank decomposition, keeping the original weights frozen while training only the new update matrices. In this study, we combined LoRA with MoE for efficient continual MLLM fine-tuning.

3. Methodology

In this section, we first present the problem formulation for multimodal continual learning. Then in Section 3.2, we introduce CoPE, a Mixture-of-Experts based framework designed to balance efficient expert expansion with anti-

forgetting capabilities. Finally, we detail the training objectives in Section 3.3.

3.1. Problem Formulation

Let $\{\mathcal{T}_1, \dots, \mathcal{T}_N\}$ be a set of N tasks, where each task $\mathcal{T}_i$ has its training set $\mathbf{D}^i$ consisting of n_i multimodal inputs $\mathbf{X} \in \{\mathbf{X}_t^i, \mathbf{X}_{img}^i, \mathbf{X}_l^i\}_{i=1}^{n_i}$. Here, $\mathbf{X}_t^i$, $\mathbf{X}_{img}^i$, and $\mathbf{X}_l^i$ represent the textual input (instruction), image, and the corresponding answer, respectively. In Continual Learning, the model is trained sequentially on the N tasks, and while training the i -th task $\mathcal{T}_i$, the model needs to maximize the probability P_i through Next Token Prediction. Notably, while training the i -th task, it cannot access to the data of previous tasks $\{\mathcal{T}_1, \dots, \mathcal{T}_{i-1}\}$. During the inference phase, we receive only task-agnostic data from either seen or unseen tasks. This practical constraint differentiates our approach from traditional task-incremental learning (Task-IL) paradigms, where explicit task identifiers are typically accessible during inference (Shi et al., 2024), thus significantly enhancing practical applicability in real-world scenarios.

3.2. CoPE

As illustrated in Figure 2, CoPE addresses the forgetting problem in MoE architectures by optimizing two key components: the experts and the router. First, we introduce the base MoE architecture in Section 3.2.1. To tackle forgetting in experts, we propose the **PGKE** mechanism (Section 3.2.2), which dynamically identifies optimal locations for expert expansion. For the router, we introduce **PTL** (Section 3.2.3) to ensure inputs are routed to task-relevant experts, thereby maximizing knowledge utilization during inference.

3.2.1. MIXTURE OF EXPERT LAYER

Our model is built upon a multimodal large language model, such as LLaVA (Liu et al., 2023), in which the MoE is implemented by augmenting the Feed-Forward Network modules within each Transformer block. Furthermore, each expert is constructed using a LoRA module, parameterized by $\{\mathbf{A}_i, \mathbf{B}_i\}_{i=1}^{N_e}$ (Hu et al., 2022), where N_e denotes the number of experts. In the h -th block, given the multimodal token $\mathbf{X}^h$, the output token $\mathbf{X}_{out}^h$ is computed as follows:

$$\mathbf{X}_{out}^h = \mathbf{W}_0 \mathbf{X}^h + \sum_{i=1}^{N_e} \omega'_i \mathbf{B}_i \mathbf{A}_i \mathbf{X}^h, \quad (1)$$

where $\omega'_i = \frac{\omega_i}{\sum_{j=1}^{N_e} \omega_j}$, $\omega_i = \exp(\mathbf{G} \mathbf{X}^h) \cdot \mathbb{I}[i \in \text{topk}(\mathbf{G} \mathbf{X}^h)]$ and $\mathbf{W}_0$, $\mathbf{G}$ represent the linear layers’ weights of the Feed-Forward Network and the router network, respectively. $\mathbb{I}$ denotes the indicator function. In our gating mechanism, the value of k for the top- k operation is not fixed. Instead, we dynamically set $k = \lfloor N/4 \rfloor$, where N is the total number of experts at the current stage, to maintain a healthy expert

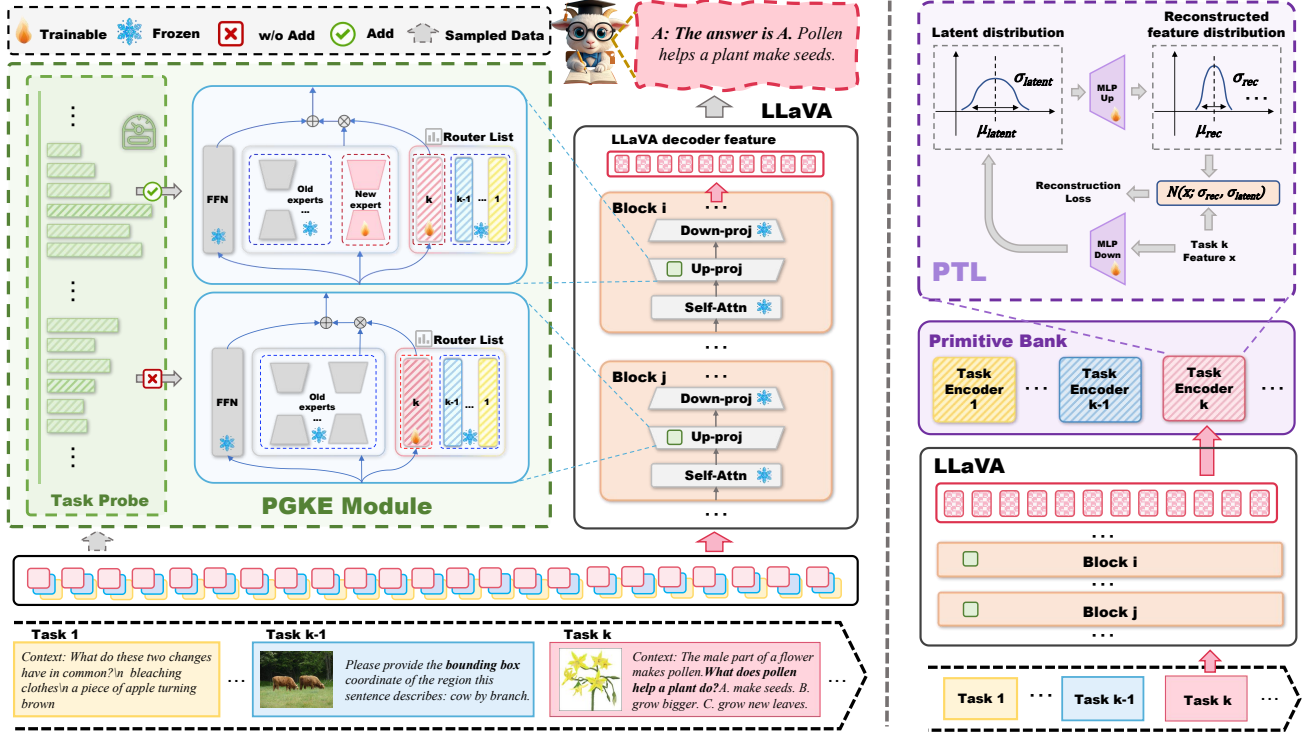


Figure 2. **Overview of our CoPE.** Our model consists of two main components: 1) **Probe-Guided Knowledge Expansion** adaptively expands experts for different tasks based on task probe guidance, enabling efficient task learning. 2) **Probabilistic Task Locator** establishes the connection between task distributions and task routing. During inference, it identifies the corresponding router based on the input, ensuring accurate task-specific processing. As illustrated in the left part, Block i was selected by PGKE to be expanded with new experts, while Block j was not. The right side shows that the model stores the features of task k into the primitive bank via PTL.

activation rate.

3.2.2. PROBE-GUIDED KNOWLEDGE EXTENSION

Current MoE-oriented continual learning methods usually append a *fixed* number of experts to *every* layer (Wang & Li, 2024; Yang et al., 2024) when training a new task. However, this simple solution (i) wastes parameters when the new task is similar to previous ones, (ii) scales quadratically with the number of tasks, and (iii) still alters the shared router, inducing forgetting. To address these issues, we propose **Probe-Guided Knowledge Extension**, **measuring where the current model lacks capacity before it allocates new experts**. The key intuition is that, if the features required by the incoming task already lie in the span of existing experts, then a freshly initialized “probe” expert will rarely be selected. Conversely, persistent high probe activations indicate that *no existing expert can explain the new examples*, indicating that extra capacity is truly needed. This “try-before-you-buy” principle keeps the parameter budget tight and aligns expansion with genuine knowledge gaps.

Following this principle, we design a two-stage training framework, consisting of *probe locating* and *expert expansion*. Given the training set $\mathbf{X}^i$ for the i -th task, we begin by sampling two non-overlapped subsets: $\mathbf{X}_{\text{train}}^i$ for training

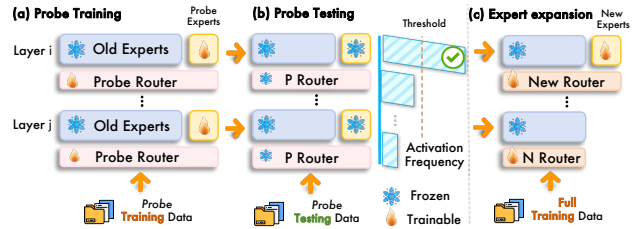


Figure 3. Illustration of Probe-Guided Knowledge Extension process. a) Probe experts are added to all layers. We use probe training data to train probe experts and probe routers. b) Calculating activation frequencies to select layers to expand. c) Expanding selected layers’ experts and doing full training.

and $\mathbf{X}_{\text{eval}}^i$ for evaluation of the probe experts.

Probe locating. In this phase, each MoE layer is augmented by introducing new probe experts along with a corresponding probe router, denoted as $\mathbf{R}_i$. To ensure accurate probing, $\mathbf{R}_i$ is initialized with the parameters of the router from the $(i - 1)$ -th task, new probe experts are initialized with the average weight of the old expert group. An additional dedicated probe router is appended to accommodate the probe experts. During this phase, the parameters of all existing experts are frozen; only the parameters of the probe experts and $\mathbf{R}_i$ are updated using $\mathbf{X}_{\text{train}}^i$.

Upon completion of probe training, we evaluate the activation statistics of all experts in each layer over the validation set $\mathbf{X}_{\text{eval}}^i$ by computing both the mean and variance of their activation frequencies (logit values). Based on these statistics, we define a threshold for expert expansion as follows (with the layer index omitted for clarity):

$$\text{Threshold} = \text{mean}(\mathbf{Act}) - \alpha \cdot \text{std}(\mathbf{Act}), \quad (2)$$

where α is a hyperparameter that modulates the sensitivity of expert growth (set to 0.8). If the activation frequencies of N_s probe experts exceed this threshold, we interpret this as evidence that the current layer requires additional capacity, and therefore augment the layer by introducing N_s new experts accordingly. Otherwise, we keep the current layer unchanged. The selection process is illustrated in Figure 3.

Expert expansion. After determining the optimal number of experts to be integrated into each layer, we perform expert expansion by augmenting the selected layers with the corresponding number of newly initialized experts and routers, following the same initialization strategy as for $\mathbf{R}_i$. Additionally, the weights of the new expert are derived from the weights of the expert that is most frequently activated during the probe locating process. Subsequently, the model is fine-tuned on the entire training dataset $\mathbf{X}^i$, freezing old experts to mitigate forgetting. To accommodate the varying complexities of different tasks, the parameter N_s , representing the number of newly added experts, is treated as a dynamically adjustable quantity. This approach not only enhances the efficiency of parameter expansion, but also ensures that the model can effectively adapt to and learn from tasks with greater complexity. A comprehensive analysis of its impact is also presented in Section 4.3.

3.2.3. PROBABILISTIC TASK LOCATOR

While PGKE effectively preserves task-specific knowledge, deploying the model without explicit task identifiers requires a mechanism to bridge input data to the correct experts. A standard shared router often suffers from catastrophic forgetting, drifting towards the most recent tasks. To resolve this without relying on replay data, we integrate a Probabilistic Task Locator to facilitate automatic router selection.

Variational Reconstruction Strategy. Instead of training a separate classifier, which may itself degrade over time, we draw inspiration from variational anomaly detection methods (Pu et al., 2016; Pinheiro Cinelli et al., 2021; An & Cho, 2015). We treat the hidden representation of each task as a unique distribution and fit a lightweight Variational Autoencoder to model it. This allows us to assess how likely a new input belongs to a previous task based on reconstruction quality, effectively converting the routing problem into a distribution matching problem.

Primitive Bank Construction. To enable efficient infer-

ence, we maintain a lightweight *Primitive Bank* that links task statistics to their corresponding PGKE modules. Specifically, upon completing the training for task i , we record the empirical mean and standard deviation of the reconstruction probabilities generated by its VAE. This bank, denoted as $\mathcal{B}$, serves as a static reference for task identity, incurring negligible storage overhead compared to traditional replay buffers. To construct our Primitive Bank, we set the sampling parameter $T = 1.0$ to accurately estimate the task-specific distributions.

Inference Process. During the inference phase, PTL computes the reconstruction probability of the incoming sample against every primitive in $\mathcal{B}$. The task index yielding the highest normalized probability is identified as the best match, automatically activating the corresponding frozen router and experts. This design ensures that our framework remains strictly replay-free while achieving high routing accuracy on unseen data. Detailed formulations and training objectives are provided in the Appendix.

3.3. Training Objective

The training loss of PTL consists of two components: the reconstruction loss $\mathcal{L}_{\text{rec}}$ and the Kullback-Leibler (KL) divergence $\mathcal{L}_{\text{KL}}$ between the posterior distribution and the prior distribution of the latent space. The reconstruction loss can be measured by the negative reconstruction probability.

We follow the common setting of VAE that the prior distribution of the latent space is $\mathcal{N}(\mathbf{0}, \mathbf{1})$ (Kingma & Welling, 2013). The posterior distribution of the latent space is $\mathcal{N}(\mu_{\text{latent}}, \sigma_{\text{latent}}^2)$. The KL divergence $\mathcal{L}_{\text{KL}}$ is calculated between the two distributions. For PGKE, we employ the next token prediction training schema and utilize a cross-entropy loss $\mathcal{L}_{\text{CE}}$. Besides, referring to (Fedus et al., 2022), we also add a weight balance loss $\mathcal{L}_{\text{LB}}$ for MoE training. Finally, the complete loss is composed of a weighted sum of these components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{KL}} + \eta \mathcal{L}_{\text{rec}} + \kappa \mathcal{L}_{\text{LB}}, \quad (3)$$

where λ , η and κ are the weights for loss balance.

4. Experiments and Discussions

4.1. Setups and Implementation Details.

Datasets and Metrics. We conducted experiments on the datasets included in the CoIN (Chen et al., 2024) benchmark, which encompasses a series of eight VQA tasks. We adopt the metric introduced in CoIN (Chen et al., 2024), which measures the discrepancy between the model’s output and the ground truth. For assessing the model’s overall forgetting performance across all tasks, we utilized Backward Transfer (BWT), which evaluates the model’s performance on all previous tasks after it is trained on the last task, specif-

Table 1. A comprehensive comparison with baseline models and other continual learning approaches built upon LLaVA is detailed in the subsequent section. *Immediate* means performance after immediate task training. *Last* means performance after the last task training. *Mean* means the average accuracy on all eight tasks.

Setting	Method	Accuracy on Each Task								Mean↑	BWT↑
		SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA		
Multitask	MoELoRA	75.01	58.90	96.44	58.15	56.73	27.54	64.04	47.81	60.58	–
Immediate	LoRA (Hu et al., 2022)	75.01	58.36	96.08	53.87	56.54	18.32	63.23	53.96	59.42	–
	MoELoRA (Luo et al., 2024)	78.97	61.01	97.01	56.24	56.30	20.63	66.10	60.57	62.10	–
	EWC (Schwarz et al., 2018)	79.23	61.26	96.91	56.43	60.04	19.21	66.00	60.44	62.44	–
	LWF (Li & Hoiem, 2017)	78.83	61.57	97.07	56.75	53.48	20.57	65.27	61.10	61.83	–
	MoExtend (Zhong et al., 2024)	77.43	59.31	96.77	56.47	57.91	24.18	64.32	60.11	62.06	–
	Ours	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81	–
Last	LoRA (Hu et al., 2022)	68.50	42.01	34.65	40.39	40.87	3.60	55.29	53.96	42.41	-17.01
	MoELoRA (Luo et al., 2024)	67.06	48.16	36.22	41.83	41.00	2.62	56.48	60.57	44.24	-17.86
	EWC (Schwarz et al., 2018)	60.25	47.92	30.36	41.33	31.12	4.53	56.31	60.44	41.53	-20.90
	LWF (Li & Hoiem, 2017)	65.15	49.46	32.52	41.05	37.88	4.81	56.20	61.10	43.51	-18.31
	O-LoRA (Wang et al., 2023a)	75.40	52.89	71.85	47.30	37.35	7.10	61.85	61.20	51.87	-17.43
	LoTA (Panda et al., 2024)	67.30	41.51	8.25	37.15	42.25	0.10	47.95	56.15	37.58	-17.14
	SEFE (Chen et al., 2025)	75.35	58.66	83.10	54.25	48.85	16.75	65.35	66.25	58.57	-10.45
	Ours	77.55	58.17	94.50	48.91	55.45	23.40	56.40	59.44	59.23	-3.58

ically quantifying the extent of forgetting, offering insights into the model’s ability to retain knowledge from previous tasks. Details can be found in the supplementary material.

Baseline Models and Methods. Following CoIN (Chen et al., 2024), we also adopted several other representative methods based on architecture and regularization, including EWC (Schwarz et al., 2018), LWF (Li & Hoiem, 2017), MoExtend (Zhong et al., 2024), O-LoRA (Wang et al., 2023a), LoTA (Panda et al., 2024), SEFE (Chen et al., 2025) as baselines to compare with our proposed method. It is important to note that all of these compared methods operate under the paradigm of training with clear task boundaries. Notably, EWC (Schwarz et al., 2018) also requires task IDs during the inference phase. We also reproduced the parameter expansion mechanism of Moextend (Zhong et al., 2024) as an “Immediate” Setting baseline. Given that it is not primarily focused on continual learning, we did not evaluate its “Last” setting. Furthermore, we align factors that could potentially affect fairness, such as the rank of LoRA and the data used for initializing the model. For more details, refer to CoIN (Chen et al., 2024).

4.2. Quantitative Results on CoIN Benchmark

As presented in Table 1, we conducted an evaluation on the CoIN (Chen et al., 2024) Benchmark. Compared with other methods, our model demonstrates outstanding performance in both immediate and post-last-task training phases, while the average number of trainable parameters in our model

is only 43.96M, more details can be found in the appendix. This significantly reduces the training cost. In the setting of *Last*, our method significantly outperforms previous approaches, demonstrating its strong capability in mitigating forgetting, especially on the ImageNet dataset, our method achieves an improvement of nearly 85%. Additionally, it is observed that on the OCR-VQA dataset, all other methods yield consistent results across both settings. However, when a new task is introduced, these methods exhibit substantial forgetting on this dataset, as evidenced by the performance in the first seven tasks. In contrast, our method maintains a stable performance of approximately 59%, showcasing its robustness against forgetting. Besides, as evident from the left of Table 2, employing our proposed method for expert extension and training yields superior results on most tasks compared to the outcomes of training without expanding experts and unfreezing the previously frozen experts.

4.3. Ablation Study and Analysis

Where should the experts be extended? First, as shown in the left of Table 2, expanding experts is vital for learning new knowledge and confronting forgetting. When expanding experts, a common practice is to add experts to every layer for each task (Yang et al., 2024). However, we argue that this approach is inefficient due to significant knowledge overlap between tasks, which leads to parameter redundancy. To validate the effectiveness of our PGKE algorithm, as shown in Table 3, we compare several expansion strategies. By comparing the results of Ours and Every-layer, it is evi-

Table 2. **Left:** Comparison of forgetting between the Extend and w/o Extend models. “Extend” freezes original experts and adds new ones using our method, while “w/o Extend” unfreezes all experts and continues training without adding new ones. **Right:** Comparison of expert addition strategies. “Ours” adds experts at selected layers using our method, “Random” adds experts randomly at the same number of layers, and “Every-layer” adds experts to all layers. Param-ratio is the proportion of parameters added by Ours compared to Every-layer.

Dataset	w/o Extend	Extend	Dataset	Random	Every-layer	Ours	Param-ratio
SQA	76.68	77.55	SQA	76.73	80.03	79.01	0.65
TQA	49.42	58.17	TQA	58.87	60.07	59.94	0.812
ImageNet	45.72	94.50	ImageNet	96.75	97.09	96.85	0.75
GQA	44.65	48.91	GQA	57.60	57.28	56.43	0.625
VizWiz	46.79	55.45	VizWiz	54.62	56.52	57.44	0.5
Ref	4.00	23.40	Ref	20.98	31.68	25.63	0.75
VQAv2	58.58	56.40	VQAv2	64.20	64.97	65.15	0.75
OCR-VQA	60.24	59.44	OCR-VQA	56.87	59.78	62.01	0.84
BWT↑	-17.68	-3.58					

Table 3. Comparison of different task orders. “Norm” denotes that we follow CoIN’s training order. “Rev” denotes that we reverse CoIN’s training order. “Rand” denotes that we random shuffle the training order of the CoIN’s datasets.

Setting	Immediate			Last		
	Norm	Rev	Rand	Norm	Rev	Rand
ScienceQA	79.01	79.96	79.63	77.55	78.42	78.48
TextVQA	59.94	58.60	58.51	58.17	56.12	56.01
ImageNet	96.85	96.99	96.89	94.50	94.61	94.55
GQA	56.43	57.88	57.81	48.91	50.32	50.29
VizWiz	57.44	58.76	57.92	55.45	56.77	56.70
Ref	25.63	29.29	29.13	23.40	27.23	27.09
VQAv2	65.15	64.75	64.83	56.40	56.02	56.16
OCR-VQA	62.01	61.27	61.31	59.44	58.72	58.66
Mean ↑	62.81	63.43	63.25	59.23	59.78	59.74
BWT ↑	-	-	-	-3.58	-3.66	-3.51

dent that our method achieves comparable performance with only 60% of the parameters, even outperforming the every-layer approach on OCR-VQA. Furthermore, the comparison between Random and Ours demonstrates that the positions identified by the task probe are more reasonable, yielding an average performance improvement of 3%-4% under the same training parameter budget, showing the superiority of the PGKE method.

How does task order influence performance? We conducted experiments on our method under different training orders across eight datasets of CoIN to verify its robustness to training order. As shown in the table, the forgetting level of our method remains stable across different training orders. This is because the feature extraction and reconstruction of different tasks are independent, same as task routers. Previously trained tasks do not affect the features of subsequent tasks. Additionally, training order has a minimal impact on the method’s immediate performance, though this is not our primary focus. More details will be discussed in the Appendix.

How many experts should be extended? Our experiments show that for certain challenging tasks, increasing the number of parameters is crucial, even when added to layers with overlapping knowledge (see the left side of Table 2). To investigate further, we examined how the number of experts per layer affects performance on the Ref and SQA tasks by adding 1, 4, or 8 experts to probe-selected layers (results in Table 6). We found that performance on the more difficult Grounding task improves significantly with more parameters, while the simpler SQA task benefits only marginally. However, due to the distinction between task difficulty and task dissimilarity, we could not develop an end-to-end method for determining the optimal number of experts based on task difficulty. As a result, we focus on optimizing layer selection and treat the number of experts as a hyperparameter.

How does the PTL mechanism perform? To figure out the PTL mechanism, we conducted the following two ablation experiments: 1) **Last** task evaluation: We utilized the router and the corresponding set of experts learned from the last task to evaluate all previous tasks. 2) **Random** task evaluation: We randomly selected a task router and its corresponding set of experts to evaluate all tasks. Results are presented in the left part of Table 7. The last task is excluded since it is not affected by forgetting. First, using either the last or the random task’s router results in catastrophic forgetting w.r.t the routers of previous tasks. In contrast, our PTL learns an easily-scalable task locator that adaptively selects tasks’ specific routers, better retains knowledge of past tasks. Additional experiments on feature extraction methods are detailed in the Appendix.

Can knowledge from previous tasks facilitate new task learning? In continual learning, beyond mitigating forgetting, it is also important to assess whether prior knowledge facilitates learning new tasks. To this end, we evaluated the model’s forward transfer by comparing it to models trained from scratch on each of the eight tasks, with an equal num-

Table 4. Performance Comparison of Different Initialization Strategies

Initialization Strategy	SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR	Mean
Zero Init	77.42	57.13	93.22	55.39	55.32	23.12	64.13	60.39	60.77
Average Init	78.33	57.94	95.37	55.25	57.72	24.62	64.30	62.16	61.96
Probe Expert Init	79.13	59.44	96.51	56.55	57.29	25.27	65.21	61.77	62.65
Ours (Nearest Expert)	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81

Table 5. Performance and Expert Addition Comparison with/without Load Balancing

Setting	Metric	SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR	Avg. Added
w/o Load Balancing	Accuracy (%)	79.05	59.89	96.65	56.32	57.52	25.75	65.18	62.22	-
	Experts Added	24	27	24	21	18	27	26	27	24.25
Ours (with LB)	Accuracy (%)	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	-
	Experts Added	21	26	24	20	16	24	24	27	22.75

Table 6. The impact of the number of experts added at specified layers on the final performance.

Number of Added Experts	1	4	8
Ref	25.63	35.13	41.51
ScienceQA	79.01	81.42	82.01

Table 7. Left: Comparison of different task classification strategies. Right: Knowledge forward transfer ability comparison.

Dataset	Last	Random	Ours	Dataset	Separate	Ours
SQA	73.03	61.92	77.55	SQA	78.97	79.01
TQA	46.82	50.62	58.17	TQA	60.56	59.94
ImageNet	29.68	36.59	94.50	ImageNet	97.05	96.85
GQA	41.81	43.12	48.91	GQA	56.31	56.43
VizWiz	44.32	37.86	55.45	VizWiz	56.20	57.44
Ref	9.92	4.83	23.40	Ref	21.3	25.63
VQAv2	55.13	45.90	56.40	VQAv2	65.01	65.15
OCR	62.01	25.62	59.44	OCR	60.79	62.01

ber of trainable parameters, as shown in the right side of Table 7. Results show that while prior knowledge offers limited benefits for simpler tasks, it significantly accelerates learning on more challenging tasks like Ref and OCR-VQA.

Initialization of New Expert. To determine the optimal strategy, we conducted a comprehensive ablation study comparing four distinct initialization methods: (1) Zero Initialization, (2) Average Initialization, (3) Initialization from Probe Experts, and (4) Initialization from the Nearest Expert (our proposed method, inheriting weights from the most semantically similar expert of the previous task).

As shown in Table 4, initializing from the nearest expert yields the superior performance on the majority of datasets (e.g., TextVQA, ImageNet, Ref, OCR-VQA). We attribute this success to a “warm start” mechanism: in the early stages of training, a new expert initialized with valid prior

knowledge (from a similar old expert) has a significantly higher probability of being selected by the router compared to a randomly or averagely initialized one. This ensures the expert receives sufficient gradient updates early on, allowing it to rapidly adapt and eventually differentiate its capabilities to fit the new task requirements. We have incorporated these comparative results into the revised manuscript to provide a rigorous justification for our design choice.

Will probe activation inflated by load-balancing loss? Theoretically, since probe experts are initialized from an average distribution without strong priors, they are at a disadvantage compared to the well-trained, frozen experts. Without the LB loss, the router is susceptible to unstable convergence, often resulting in a “winner-take-all” scenario or inefficient routing decisions. To empirically verify this, we conducted an ablation study removing the Load Balancing loss during the probe stage. As shown in the Table 5, removing the LB loss actually results in a higher rate of expert expansion (e.g., adding 24 experts vs. 21 on SQA, and 18 vs. 16 on VizWiz) without yielding significant performance gains. This indicates that without the regularization provided by the LB loss, the router tends to blindly allocate new parameters. Conversely, the inclusion of LB loss effectively “rationalizes” the selection process; it prevents unnecessary expansion by enforcing a fairer probability distribution, which, in practice, encourages the model to leverage the capabilities of existing experts rather than defaulting to new ones. Thus, the LB loss serves as a critical regularizer to minimize redundant parameter growth while maintaining optimal performance.

5. Conclusion and Limitation

In this paper, we propose CoPE, a continual learning framework comprising two key modules: Probe-Guided Knowledge Extension and Probabilistic Task Locator. PGKE addresses the inefficiency of continuous parameter expansion

by adaptively increasing parameters through probe-guided expert addition. Meanwhile, PTL mitigates catastrophic forgetting in continual learning by modeling task distributions and memorizing the mapping between task distributions and router networks.

Limitation and Future Work. Currently, expert addition is restricted to the language model, potentially limiting performance on tasks requiring fine-grained visual understanding. Additionally, key parameters such as the number of new experts (N_s) and the layer selection threshold (α) are currently set empirically. Furthermore, storage demands grow with the number of tasks, and standard compression techniques may degrade router performance. In future work, we aim to address these challenges by developing task-adaptive capacity controllers and automated hyperparameter optimization frameworks to make N_s end-to-end learnable, which is crucial for handling extreme data volume imbalances across task streams. We also plan to explore extending our probe-guided expansion mechanism to the visual module to further enhance fine-grained visual comprehension

Acknowledgements

This research is supported in part by the National Natural Science Foundation of China (Nos. 62461160308, U23B2010, 62576024, L231011), the "Pioneer" and "Leading Goose" R&D Program of Zhejiang (No. 2024C01161), the Ningbo Science and Technology Innovation 2025 Major Project (No. 2025Z034), and the Key Research Program of Hangzhou (No. 2025SZD1A56). In addition, this work was supported by the Fundamental Research Funds for the Central Universities (No. 501RCQD2025141003), the Bei-Hang GanWei Project (No. 502GWXM2024141001), and the NUS Start-up Grant (No. A-0010106-00-00).

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Abou Baker, N., Rohrschneider, D., and Handmann, U. Parameter-efficient fine-tuning of large pretrained models for instance segmentation tasks. *Machine Learning and Knowledge Extraction*, 6(4):2783–2807, 2024.
- An, J. and Cho, S. Variational autoencoder based anomaly detection using reconstruction probability, 2015. URL <https://api.semanticscholar.org/CorpusID:36663713>.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33: 1877–1901, 2020.
- Caffagni, D., Cocchi, F., Barsellotti, L., Moratelli, N., Sarto, S., Baraldi, L., Baraldi, L., Cornia, M., and Cucchiara, R. The revolution of multimodal large language models: A survey. In Ku, L.-W., Martins, A., and Srikumar, V. (eds.), *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 13590–13618, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.807. URL <https://aclanthology.org/2024.findings-acl.807/>.
- Cai, J., Wang, X., Guan, C., Tang, Y., Xu, J., Zhong, B., and Zhu, W. Multimodal continual graph learning with neural architecture search. In *Proceedings of the ACM Web Conference 2022*, pp. 1292–1300, 2022.
- Cai, Y., Thomason, J., and Rostami, M. Task-attentive transformer architecture for continual learning of vision-and-language tasks using knowledge distillation. *arXiv preprint arXiv:2303.14423*, 2023.
- Chen, C., Zhu, J., Luo, X., Shen, H., Song, J., and Gao, L. Coin: A benchmark of continual instruction tuning for multimodal large language models. *Advances in Neural Information Processing Systems*, 37:57817–57840, 2024.
- Chen, J., Cong, R., Zhao, Y., Yang, H., Hu, G., Ip, H. H. S., and Kwong, S. Sefe: Superficial and essential forgetting eliminator for multimodal continual instruction tuning. *arXiv preprint arXiv:2505.02486*, 2025.
- Chen, W., Zhou, Y., Du, N., Huang, Y., Laudon, J., Chen, Z., and Cui, C. Lifelong language pretraining with distribution-specialized experts. In *International Conference on Machine Learning*, pp. 5383–5395. PMLR, 2023a.
- Chen, Y., Hazarika, D., Namazifar, M., Liu, Y., Jin, D., and Hakkani-Tur, D. Empowering parameter-efficient transfer learning by recognizing the kernel structure in self-attention. In *Findings of the Association for Computational Linguistics: NAACL 2022*, pp. 1375–1388, 2022.
- Chen, Z., Wang, Z., Wang, Z., Liu, H., Yin, Z., Liu, S., Sheng, L., Ouyang, W., Qiao, Y., and Shao, J. Octavius: Mitigating task interference in mllms via lora-moe. *arXiv preprint arXiv:2311.02684*, 2023b.
- Chiaro, R. D., Twardowski, B., Bagdanov, A. D., and van de Weijer, J. Ratt: Recurrent attention to transient tasks

- for continual image captioning, 2020. URL <https://arxiv.org/abs/2007.06271>.
- Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H. W., Sutton, C., Gehrmann, S., et al. Palm: Scaling language modeling with pathways. *Journal of Machine Learning Research*, 24(240):1–113, 2023.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.
- D’Alessandro, M., Alonso, A., Calabrés, E., and Galar, M. Multimodal parameter-efficient few-shot class incremental learning. In *2023 IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, pp. 3385–3395. IEEE, October 2023. doi: 10.1109/iccvw60793.2023.00364. URL <http://dx.doi.org/10.1109/ICCVW60793.2023.00364>.
- Edalati, A., Tahaei, M., Kobzyev, I., Nia, V. P., Clark, J. J., and Rezagholizadeh, M. Krona: Parameter efficient tuning with kronecker adapter. *arXiv preprint arXiv:2212.10650*, 2022.
- Farajtabar, M., Azizan, N., Mott, A., and Li, A. Orthogonal gradient descent for continual learning. In *International conference on artificial intelligence and statistics*, pp. 3762–3773. PMLR, 2020.
- Fedus, W., Zoph, B., and Shazeer, N. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Gurari, D., Li, Q., Stangl, A. J., Guo, A., Lin, C., Grauman, K., Luo, J., and Bigham, J. P. VizWiz Grand Challenge: Answering Visual Questions from Blind People. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3608–3617, Los Alamitos, CA, USA, June 2018. IEEE Computer Society. doi: 10.1109/CVPR.2018.00380. URL <https://doi.ieeecomputersociety.org/10.1109/CVPR.2018.00380>.
- Han, Z., Gao, C., Liu, J., Zhang, J., and Zhang, S. Q. Parameter-efficient fine-tuning for large models: A comprehensive survey, 2024. URL <https://arxiv.org/abs/2403.14608>.
- Hassabis, D., Kumaran, D., Summerfield, C., and Botvinick, M. Neuroscience-inspired artificial intelligence. *Neuron*, 95(2):245–258, 2017. ISSN 0896-6273. doi: <https://doi.org/10.1016/j.neuron.2017.06.011>. URL <https://www.sciencedirect.com/science/article/pii/S0896627317305093>.
- He, J., Guo, H., Tang, M., and Wang, J. Continual instruction tuning for large multimodal models, 2023. URL <https://arxiv.org/abs/2311.16206>.
- Houlsby, N., Giurghi, A., Jastrzebski, S., Morrone, B., De Laroussilhe, Q., Gesmundo, A., Attariyan, M., and Gelly, S. Parameter-efficient transfer learning for nlp. In *International Conference on Machine Learning*, pp. 2790–2799. PMLR, 2019.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=nZeVKeeFYf9>.
- Jha, S., Gong, D., and Yao, L. Clap4clip: Continual learning with probabilistic finetuning for vision-language models, 2024. URL <https://arxiv.org/abs/2403.19137>.
- Jiang, A. Q., Sablayrolles, A., Roux, A., Mensch, A., Savary, B., Bamford, C., Chaplot, D. S., de las Casas, D., Hanna, E. B., Bressand, F., Lengyel, G., Bour, G., Lample, G., Lavaud, L. R., Saulnier, L., Lachaux, M.-A., Stock, P., Subramanian, S., Yang, S., Antoniak, S., Scao, T. L., Gervet, T., Lavril, T., Wang, T., Lacroix, T., and Sayed, W. E. Mixtral of experts, 2024. URL <https://arxiv.org/abs/2401.04088>.
- Kazemzadeh, S., Ordonez, V., Matten, M., and Berg, T. ReferItGame: Referring to objects in photographs of natural scenes. In Moschitti, A., Pang, B., and Daelemans, W. (eds.), *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 787–798, Doha, Qatar, October 2014. Association for Computational Linguistics. doi: 10.3115/v1/D14-1086. URL <https://aclanthology.org/D14-1086/>.
- Kingma, D. P. and Welling, M. Auto-encoding variational bayes. *CoRR*, abs/1312.6114, 2013. URL <https://api.semanticscholar.org/CorpusID:216078090>.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., and Hadsell, R. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, March 2017. ISSN 1091-6490. doi: 10.1073/pnas.1611835114. URL <http://dx.doi.org/10.1073/pnas.1611835114>.

- Lao, M., Pu, N., Liu, Y., Zhong, Z., Bakker, E. M., Sebe, N., and Lew, M. S. Multi-domain lifelong visual question answering via self-critical distillation. In *Proceedings of the 31st ACM International Conference on Multimedia*, MM '23, pp. 4747–4758, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701085. doi: 10.1145/3581783.3612121. URL <https://doi.org/10.1145/3581783.3612121>.
- Lei, S. W., Gao, D., Wu, J. Z., Wang, Y., Liu, W., Zhang, M., and Shou, M. Z. Symbolic replay: Scene graph as prompt for continual learning on vqa task. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pp. 1250–1259, 2023.
- Lepikhin, D., Lee, H., Xu, Y., Chen, D., Firat, O., Huang, Y., Krikun, M., Shazeer, N., and Chen, Z. Gshard: Scaling giant models with conditional computation and automatic sharding, 2020. URL <https://arxiv.org/abs/2006.16668>.
- Li, J., Li, D., Savarese, S., and Hoi, S. C. H. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International Conference on Machine Learning*, 2023. URL <https://api.semanticscholar.org/CorpusID:256390509>.
- Li, X. L. and Liang, P. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*, 2021.
- Li, Z. and Hoiem, D. Learning without forgetting. *IEEE transactions on pattern analysis and machine intelligence*, 40(12):2935–2947, 2017.
- Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36:34892–34916, 2023.
- Liu, Q., Wu, X., Zhao, X., Zhu, Y., Xu, D., Tian, F., and Zheng, Y. When moe meets llms: Parameter efficient fine-tuning for multi-task medical applications. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 1104–1114, 2024.
- Liu, X., Ji, K., Fu, Y., Tam, W. L., Du, Z., Yang, Z., and Tang, J. P-tuning v2: Prompt tuning can be comparable to fine-tuning universally across scales and tasks. *arXiv preprint arXiv:2110.07602*, 2021.
- Liu, X., Ji, K., Fu, Y., Tam, W., Du, Z., Yang, Z., and Tang, J. P-tuning: Prompt tuning can be comparable to fine-tuning across scales and tasks. In Muresan, S., Nakov, P., and Villavicencio, A. (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 61–68, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-short.8. URL <https://aclanthology.org/2022.acl-short.8>.
- Lopez-Paz, D. and Ranzato, M. Gradient episodic memory for continual learning. *Advances in neural information processing systems*, 30, 2017.
- Luo, T., Lei, J., Lei, F., Liu, W., He, S., Zhao, J., and Liu, K. Moelora: Contrastive learning guided mixture of experts on parameter-efficient fine-tuning for large language models. *ArXiv*, abs/2402.12851, 2024. URL <https://api.semanticscholar.org/CorpusID:267759827>.
- Mustafa, B., Riquelme, C., Puigcerver, J., Jenatton, R., and Houlsby, N. Multimodal contrastive learning with limoe: the language-image mixture of experts. *Advances in Neural Information Processing Systems*, 35:9564–9576, 2022.
- Panda, A., Isik, B., Qi, X., Koyejo, S., Weissman, T., and Mittal, P. Lottery ticket adaptation: Mitigating destructive interference in llms. *arXiv preprint arXiv:2406.16797*, 2024.
- Peng, Y., Qi, J., Ye, Z., and Zhuo, Y. Hierarchical visual-textual knowledge distillation for life-long correlation learning. *Int. J. Comput. Vision*, 129(4): 921–941, April 2021. ISSN 0920-5691. doi: 10.1007/s11263-020-01392-1. URL <https://doi.org/10.1007/s11263-020-01392-1>.
- Pinheiro Cinelli, L., Araújo Marins, M., Barros da Silva, E. A., and Lima Netto, S. Variational autoencoder. In *Variational methods for machine learning with applications to deep networks*, pp. 111–149. Springer, 2021.
- Pu, Y., Gan, Z., Henao, R., Yuan, X., Li, C., Stevens, A., and Carin, L. Variational autoencoder for deep learning of images, labels and captions. *Advances in neural information processing systems*, 29, 2016.
- Qian, Z., Wang, X., Duan, X., Qin, P., Li, Y., and Zhu, W. Decouple before interact: Multi-modal prompt learning for continual visual question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 2953–2962, October 2023.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021. URL <https://api.semanticscholar.org/CorpusID:231591445>.

- Riquelme, C., Puigcerver, J., Mustafa, B., Neumann, M., Jenatton, R., Susano Pinto, A., Keysers, D., and Houlsby, N. Scaling vision with sparse mixture of experts. *Advances in Neural Information Processing Systems*, 34: 8583–8595, 2021.
- Saikh, T., Ghosal, T., Mittal, A., Ekbal, A., and Bhat-tacharyya, P. Scienceqa: A novel resource for question answering on scholarly articles. *International Journal on Digital Libraries*, 23(3):289–301, 2022.
- Schwarz, J., Czarnecki, W., Luketina, J., Grabska-Barwinska, A., Teh, Y. W., Pascanu, R., and Hadsell, R. Progress & compress: A scalable framework for continual learning. In *International conference on machine learning*, pp. 4528–4537. PMLR, 2018.
- Serrà, J., Surís, D., Miron, M., and Karatzoglou, A. Overcoming catastrophic forgetting with hard attention to the task, 2018. URL <https://arxiv.org/abs/1801.01423>.
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., and Dean, J. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv: Learning*, Jan 2017.
- Shen, S., Yao, Z., Li, C., Darrell, T., Keutzer, K., and He, Y. Scaling vision-language models with sparse mixture of experts. *arXiv preprint arXiv:2303.07226*, 2023.
- Shi, H., Xu, Z., Wang, H., Qin, W., Wang, W., Wang, Y., Wang, Z., Ebrahimi, S., and Wang, H. Continual learning of large language models: A comprehensive survey. *arXiv preprint arXiv:2404.16789*, 2024.
- Singh, A., Natarajan, V., Shah, M., Jiang, Y., Chen, X., Batra, D., Parikh, D., and Rohrbach, M. Towards VQA models that can read. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pp. 8317–8326. Computer Vision Foundation / IEEE, 2019. doi: 10.1109/CVPR.2019.00851. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Singh_Towards_VQA_Models_That_Can_Read_CVPR_2019_paper.html.
- Sun, F., Liu, H., Yang, C., and Fang, B. Multimodal continual learning using online dictionary updating. *IEEE Transactions on Cognitive and Developmental Systems*, 13(1):171–178, 2021. doi: 10.1109/TCDS.2020.2973280.
- Sun, Q., Wang, Z., Zhao, H., Li, Y., Song, K., Liu, S., Hu, X., Guo, Q., and Huang, L. Vaccinerag: Boosting multimodal large language models’ immunity to harmful rag samples. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 9189–9197, 2026.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Wang, R. and Li, P. Lemoe: Advanced mixture of experts adaptor for lifelong model editing of large language models. *arXiv preprint arXiv:2406.20030*, 2024.
- Wang, X., Chen, T., Ge, Q., Xia, H., Bao, R., Zheng, R., Zhang, Q., Gui, T., and Huang, X.-J. Orthogonal subspace learning for language model continual learning. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 10658–10671, 2023a.
- Wang, Y., Huang, Z., and Hong, X. S-prompts learning with pre-trained transformers: An occam’s razor for domain incremental learning, 2023b. URL <https://arxiv.org/abs/2207.12819>.
- Wu, J., Gan, W., Chen, Z., Wan, S., and Yu, P. S. Multimodal Large Language Models: A Survey. In *2023 IEEE International Conference on Big Data (BigData)*, pp. 2247–2256, Los Alamitos, CA, USA, December 2023. IEEE Computer Society. doi: 10.1109/BigData59044.2023.10386743. URL <https://doi.ieeecomputersociety.org/10.1109/BigData59044.2023.10386743>.
- Wu, T., Luo, L., Li, Y.-F., Pan, S., Vu, T.-T., and Haffari, G. Continual learning for large language models: A survey, 2024. URL <https://arxiv.org/abs/2402.01364>.
- Yang, R., Wang, S., Zhang, H., Xu, S., Guo, Y., Ye, X., Hou, B., and Jiao, L. Knowledge decomposition and replay: A novel cross-modal image-text retrieval continual learning method. In *Proceedings of the 31st ACM International Conference on Multimedia, MM ’23*, pp. 6510–6519, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701085. doi: 10.1145/3581783.3612207. URL <https://doi.org/10.1145/3581783.3612207>.
- Yang, S., Ali, M. A., Wang, C.-L., Hu, L., and Wang, D. Moral: Moe augmented lora for llms’ lifelong learning, 2024. URL <https://arxiv.org/abs/2402.11260>.
- Yu, J., Zhuge, Y., Zhang, L., Hu, P., Wang, D., Lu, H., and He, Y. Boosting continual learning of vision-language models via mixture-of-experts adapters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 23219–23230, 2024.

- Zhang, Q., Chen, M., Bukharin, A., He, P., Cheng, Y., Chen, W., and Zhao, T. Adaptive budget allocation for parameter-efficient fine-tuning. In *The Eleventh International Conference on Learning Representations*, 2022.
- Zhang, X., Wu, C., Zhao, Z., Lin, W., Zhang, Y., Wang, Y., and Xie, W. Pmc-vqa: Visual instruction tuning for medical visual question answering. *arXiv preprint arXiv:2305.10415*, 2023.
- Zheng, J., Ma, Q., Liu, Z., Wu, B., and Feng, H. Beyond anti-forgetting: Multimodal continual instruction tuning with positive forward transfer, 2024. URL <https://arxiv.org/abs/2401.09181>.
- Zheng, Z., Ma, M., Wang, K., Qin, Z., Yue, X., and You, Y. Preventing zero-shot transfer degradation in continual learning of vision-language models. *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 19068–19079, 2023. URL <https://api.semanticscholar.org/CorpusID:257496481>.
- Zhong, S., Gao, S., Huang, Z., Wen, W., Zitnik, M., and Zhou, P. Moextend: Tuning new experts for modality and task extension. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*, 2024.
- Zhu, Y., Feng, J., Zhao, C., Wang, M., and Li, L. Counter-interference adapter for multilingual machine translation. *arXiv preprint arXiv:2104.08154*, 2021.

A. Technical Appendices and Supplementary Material

A.1. Experimental Setup and Training Details

Datasets. We conducted experiments on the datasets included in the CoIN (Chen et al., 2024) benchmark, which encompasses a series of eight VQA tasks. These tasks include RefCOCO (Ref) (Kazemzadeh et al., 2014), ImageNet (Deng et al., 2009), TextVQA (TQA) (Singh et al., 2019), VizWiz (Gurari et al., 2018), SciencQA (SQA) (Saikh et al., 2022), among others. Each task varies in terms of the number of data samples, stylistic features, and domain characteristics. The training set comprises a total of 569k samples, while the testing set contains 261k samples.

Metrics. We adopt the metric introduced in CoIN (Chen et al., 2024), which measures the discrepancy between the model’s output and the ground truth. For assessing the model’s overall forgetting performance across all tasks, we utilized Backward Transfer (BWT), which evaluates the model’s performance on all previous tasks after it is trained on the last task, specifically quantifies the extent of forgetting, offering insights into the model’s ability to retain knowledge from previous tasks.

Model Initialization. We adopted LLaVA v1.5 that had only undergone pre-training without supervised fine-tuning as the base model. To train the initial eight experts of the model, we accurately removed the overlapping parts between the 665k fine-tuning dataset of LLaVA v1.5 and the CoIN dataset, and trained the eight experts in each layer of the model on the remaining data (approximately 150k). In addition, the visual projector of the model remained frozen throughout all training stages mentioned in this paper.

Training Environment and Hyperparameters. We train the model using $8 \times \text{H20 GPUs}$. It takes 2 hours to train the initial eight experts, followed by 15 hours for the continual learning process. Throughout the training, we set the warmup ratio to 0.03, use the AdamW optimizer, and employ torch BF16 precision with DeepSpeed stage zero2. The rank of LoRA experts is set to 64, the rank alpha to 128, and the global batch size to 128. We assign a weight of $1\text{e-}3$ to the load balancing loss of the router, the KL divergence loss, and the reconstruction loss. For training the normal experts, we use a learning rate of $2\text{e-}4$, and during the probe experts training, we increase the learning rate to $3\text{e-}4$.

Time Consumption. We present statistics on the training and inference time of the task probe in Table 8. It can be observed that, on 8 H20 GPUs, each task probe process only accounts for 15% of the model training time, which falls within an acceptable time range.

Table 8. Time consumption of each process.

Tasks	Probe training(min)	Probe test(min)	Task training(min)
SQA	4	1.5	25
TQA	7	2.6	41
ImageNet	15	2.1	113
GQA	23	2.8	174
VizWiz	6	1.8	30
Ref	21	3	169
VQAv2	22	3.1	175
OCR-VQA	25	1.8	182

A.2. Details of Probabilistic Task Locator

A.2.1. MECHANISM AND FORMULATION

As mentioned in Sec 3.2.3, PTL treats the hidden representation of multimodal inputs as samples from task-specific distributions. We fit each task distribution with a lightweight VAE. The training objective involves minimizing the reconstruction loss and the KL divergence between the posterior and prior distributions of the latent space. The reconstruction loss is measured by the negative reconstruction probability:

$$\mathcal{L}_{rec} = -p_{rec}(F_{end})$$

where F_{end} represents the feature representation. The total loss function combines this with the cross-entropy loss and auxiliary losses used in PGKE (see Eq. 4 in the main text).

Algorithm of Probe-Guided Knowledge Extension. The pseudo-code of our algorithm is shown in Algorithm 1.

Algorithm 1: Probing and Training Task $\mathcal{T}_i$

Input: Original model π , probing training data D_p^{train} , probing eval data D_p^{eval} , full training data D^{train} , threshold coefficient α

```

 $\pi_0 \leftarrow \text{clone}(\pi)$ 
 $\text{probe\_layers} \leftarrow \text{range}(N_{\text{layer}})$ 
 $\text{append\_expert}(\pi_0, \text{probe\_layers})$ 
 $\pi_0 \leftarrow \text{train}(\pi_0, D_p^{\text{train}})$ 
 $\text{prob\_freq} \leftarrow \text{compute\_per\_layer\_freq}(\pi_0, D_p^{\text{test}})$ 
for  $\text{layer} \leftarrow 0$  to  $\text{prob\_layers} - 1$  do
     $\mu \leftarrow \text{mean}(\text{prob\_freq}[\text{layer}])$ 
     $\sigma \leftarrow \text{std}(\text{prob\_freq}[\text{layer}])$ 
     $\text{freq} \leftarrow \text{prob\_freq}[\text{layer}][1]$ 
    if  $\text{freq} > \mu - \alpha \cdot \sigma$  then
         $\text{selected\_layer.append}(\text{layer})$ 
end
 $\text{append\_expert}(\pi, \text{selected\_layer})$ 
 $\pi \leftarrow \text{train}(\pi, D^{\text{train}})$ 
    
```

A.2.2. PRIMITIVE BANK CONSTRUCTION AND INFERENCE

After the VAE converges for task i , we collect the most recent T training instances to compute their reconstruction probabilities $p_{\text{rec}}(F_{end})$. We record the empirical mean and standard deviation to form a "primitive distribution" for task i . The Primitive Bank is defined as $\mathcal{B} = \{(\text{primitive}_i, \text{router}_i) | i = 1, \dots, N\}$, linking the statistical distribution to the frozen router produced by PGKE. During the task-agnostic inference phase, for an input with feature F_{end} , we evaluate and z-score normalize its reconstruction probability under every primitive in $\mathcal{B}$, yielding scores $\{\hat{p}^{(i)}\}_{i=1}^N$. The inferred task identity i^* is determined by:

$$i^* = \arg \max_i \hat{p}^{(i)}$$

Subsequently, the corresponding $\text{router}_{(i^*)}$ is activated for processing.

A.3. Comprehensive Quantitative Analysis

Sequential Performance Evolution. We evaluate our methods' performance during the entire training process. As shown in Table 9, our approach maintains excellent anti-forgetting capability after training on sequential tasks, particularly on the SQA, TQA, ImageNet, VizWiz, and Ref datasets. By comparison, PTL exhibits slightly weaker performance on the VQAv2 and GQA datasets due to a slight classification confusion issue. Nevertheless, its performance remains comparable to or better than existing methods, as demonstrated in our main manuscript.

Comparison with Replay Methods. To ensure a more rigorous comparison, we extended the baseline method, where a specialist is added to all layers for each new task, with a replay-based scheme. Specifically, when learning the N th task, we sample 10% of the data from all previous tasks (1 to $N-1$) and mix it with the current task's data for training. The experimental results are shown in Table 10. It can be observed that after incorporating additional data for training, the replay-based scheme does further improve model performance. However, it also indeed increases both data storage overhead and computational overhead.

Comparison with MoExtend. The comparative results in Table 11 demonstrate that our method achieves superior performance on the majority of datasets (e.g., SQA, TextVQA, VQAv2, OCR-VQA) compared to Moextend (Zhong et al., 2024). Crucially, our approach offers significant advantages in parameter efficiency through its adaptive allocation strategy. Unlike Moextend, which adds a fixed number of 23 experts across all tasks, our method dynamically adjusts the number

Table 9. The performance on eight datasets during sequential training. Training order: SQA → TQA → ImageNet → GQA → VizWiz → Ref → VQAv2 → OCR-VQA.

Training Dataset	Accuracy on Each Task							
	SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA
SQA	79.01	-	-	-	-	-	-	-
TQA	79.01	59.94	-	-	-	-	-	-
ImageNet	79.01	59.71	96.85	-	-	-	-	-
GQA	79.00	59.28	96.31	56.43	-	-	-	-
VizWiz	78.21	58.59	95.13	55.86	57.44	-	-	-
Ref	78.21	58.58	95.12	54.40	57.42	25.63	-	-
VQAv2	77.73	58.31	94.51	49.86	56.31	24.02	65.15	-
OCR-VQA	77.55	58.17	94.50	48.91	55.45	23.40	56.40	62.01

Table 10. Comparison with replay methods.

Setting	replayed	Accuracy on Each Task								Mean↑	BWT↑	Training-time
		SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA			
Immediate	Yes	79.84	60.17	97.02	57.27	56.59	32.11	64.88	61.92	63.73	-	20h
	No (ours)	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81	-	17h
Last	Yes	77.93	58.11	94.16	50.32	53.52	28.04	60.81	60.48	62.81	-3.36	20h
	No (ours)	77.55	58.17	94.50	48.91	55.45	23.40	56.40	59.44	59.23	-3.58	17h

of experts (ranging from 16 to 27) based on task complexity. For instance, on the VizWiz dataset, our method achieves comparable performance while adding significantly fewer experts (16 vs. 23), effectively minimizing the computational burden. We are committed to open-sourcing both our model and the reproduced baseline code to serve as a valuable resource for the community.

Table 11. Comparison of performance and parameter efficiency between Moextend and our method

Dataset	Moextend (Acc / Experts)	Ours (Acc / Experts)	Δ Performance
SQA	77.43 / 23	79.01 / 21	+1.58
TextVQA	59.31 / 23	59.94 / 26	+0.63
ImageNet	96.77 / 23	96.85 / 24	+0.08
GQA	56.47 / 23	56.43 / 20	-0.04
VizWiz	57.91 / 23	57.44 / 16	-0.47
Ref	24.18 / 23	25.63 / 24	+1.45
VQAv2	64.32 / 23	65.15 / 24	+0.83
OCR-VQA	60.11 / 23	62.01 / 27	+1.90

Additionally, to address the efficacy of layer selection, we integrated MoExtend’s expansion logic into our unified codebase and conducted a comparative analysis on the GQA → VizWiz continual learning transition. The results, detailed in Table 21, reveal a 68% overlap in the layers selected for expansion by both methods (specifically layers 0, 1, 7, 9, 18, 21, 22, 23, 24, 26, 29), indicating a broad consensus on where the architecture requires capacity growth. However, a critical divergence occurs in the remaining 32% of the layers. Our method achieves a higher accuracy on VizWiz (57.44% vs. 56.82%) despite expanding a similar number of layers. This performance gap suggests that our selection strategy identifies a more optimal set of layers for parameter allocation (e.g., prioritizing layers 3, 12, 13, 17, 27). Conversely, it implies that MoExtend may trigger expansion in layers where the task preference difference is marginal or non-critical (e.g., layers 2, 14, 28, 30, 31),

Table 12. Comparison of the average trainable parameters (M) over eight datasets, and performance between LoRA, MoELoRA, EWC, LWF and Ours.

Metrics	LoRA	MoELoRA	EWC	LWF	Ours
#Trainable Parameters per Task	31M	62M	62M	62M	43.96M
Mean $\uparrow$	42.41	44.24	41.53	43.51	59.23
BWT $\uparrow$	-17.01	-17.86	-20.90	-18.31	-3.58

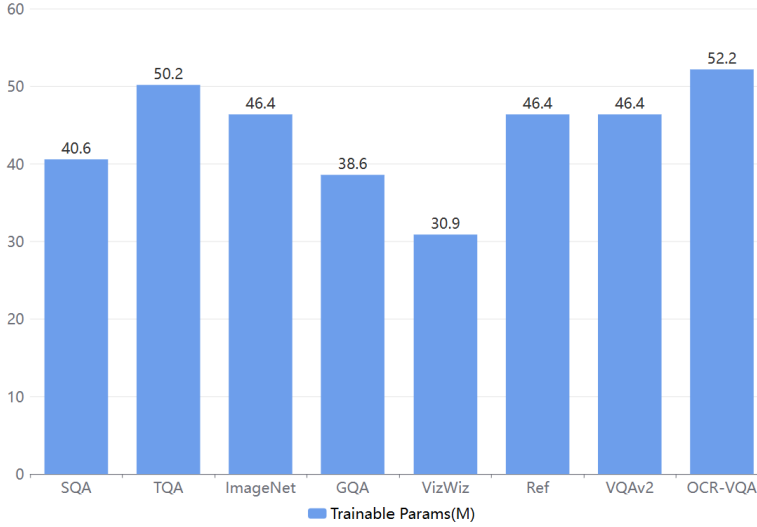


Figure 4. The added trainable parameters on eight tasks.

potentially leading to less efficient resource utilization.

Parameter Comparison. We tabulated the number of parameters for different methods, as shown in Table 12. It is worth noting that the LoRA method we compared only adds LoRA matrices to the up-proj layer. Although the LoRA method has the fewest trainable parameters, it exhibits poor anti-forgetting performance. Our method adaptively adds experts based on task differences, significantly enhancing anti-forgetting capabilities while maintaining minimal parameter growth. The changes in trainable parameters across different tasks are illustrated in Figure 4.

Expansion of experts per layer over tasks. For each task, we recorded the layers expanded during each extension in Table 13. Since the number of expanded layers exceeds that of non-expanded layers, for convenience, we list the non-expanded layers for each task: It can be observed that the model adds significantly fewer experts to the first 16 layers (low-level features) and predominantly expands parameters in the last 16 layers (high-level features). This aligns with our intuition, as the model shares some low-level features and mainly adds high-level knowledge.

Expert diversity. To verify the capacity differentiation of the newly added experts, we randomly selected all experts from one layer of the model and computed their cosine similarity. The results are presented in the Table 14. Notably, aside from the eight initial experts (with IDs 1 to 8), the newly added experts exhibit the highest similarity to their source experts (the ones they were duplicated from), with an average of approximately 0.6. This phenomenon can be attributed to the fact that during the probing phase, the expert with the highest activation is not only the closest to the new task distribution within the current expert subgroup but also the most competent in handling the new task. Consequently, the newly duplicated experts retain high similarity to the original experts after fitting the new task.

However, the most critical finding is the shift in similarity from 1.0 (at initialization) down to 0.6 after training. This significant divergence, when viewed alongside the substantial performance gains on the new task, serves as compelling evidence that the new experts have not merely retained their initialization weights but have undergone substantial gradient

Table 13. The scaling of per-layer experts as tasks expand.

Tasks	Layer ids not expand expert
SQA	1, 2, 3, 4, 6, 9, 10, 14, 15, 23, 24
TQA	0, 2, 3, 4, 5, 29
ImageNet	0, 1, 3, 6, 7, 8, 19, 21
GQA	0, 2, 3, 6, 7, 10, 11, 13, 15, 24, 25, 27
VizWiz	2, 4, 5, 6, 8, 10, 11, 14, 15, 16, 19, 20, 25, 28, 30, 31
Ref	1, 4, 8, 12, 14, 17, 23, 28
VQAv2	3, 4, 5, 12, 16, 23, 27, 31
OCR-VQA	1, 7, 1, 14, 27

Table 14. Expert diversity analysis.

Expert id	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	1	0.021	0.019	0.022	0.017	0.023	0.016	0.017	0.017	0.019	0.019	0.019	0.009	0.008
2		1	0.011	0.014	0.020	0.017	0.016	0.018	0.013	0.022	0.015	0.019	0.017	0.027
3			1	0.015	0.013	0.019	0.021	0.020	0.012	0.021	0.020	0.701	0.014	0.006
4				1	0.012	0.009	0.003	0.019	0.011	0.021	0.038	0.010	0.018	0.032
5					1	0.024	0.008	0.016	0.006	0.011	0.008	0.014	0.646	0.603
6						1	0.016	0.024	0.648	0.025	0.017	0.032	0.029	0.021
7							1	0.021	0.022	0.687	0.602	0.022	0.021	0.007
8								1	0.022	0.063	0.675	0.023	0.026	0.033
9									1	0.036	0.045	0.091	0.017	0.009
10										1	0.017	0.029	0.028	0.035
11											1	0.032	0.017	0.014
12												1	0.016	0.018
13													1	0.635
14														1

updates. This confirms that the added experts have effectively differentiated their capabilities, specializing to capture the specific features of the new domain while benefiting from a "warm start."

A.4. Ablation Studies

Analysis of PTL Mechanism and Confusion Matrix. We evaluated the PTL mechanism on test data across eight tasks, and the results are shown in Figure 5. As observed from the test results, the PTL mechanism achieves localization accuracies exceeding 80% on five tasks: ScienceQA, ImageNet, VizWiz, Grounding and OCR-VQA. Besides, we can also find that the localization performance on the remaining three tasks, GQA, TextVQA, and VQAv2, is relatively weaker compared to the other five tasks. Through our analysis, this is attributed to the significant overlap in the features of images and questions across these tasks. For instance, GQA and VQAv2 share high similarities in terms of question formats, image content, and styles, which leads to a substantial portion of VQAv2 samples being "mislocalized" to the GQA task.

Impact of selection threshold α in PGKE. We conducted an ablation experiment on the threshold α in PGKE, and the results are shown in Table 15. As observed from the results, with a strict threshold (e.g., $\alpha = 0.4$), while the average parameter growth is reduced, the network exhibits a slight performance decline in the early training stages. Particularly, in the later training stages, after the network has accumulated substantial knowledge, a strict threshold increases the difficulty for the network to capture the distributional differences between the current task and previously accumulated knowledge. This causes the network to tend to avoid adding new parameters, resulting in significant performance degradation on tasks such as OCR-VQA and VQAv2. Similarly, when the threshold is set to a loose value (e.g., $\alpha = 1.2$), the model's performance approaches that of the every-layer setting, and no substantial parameter savings are achieved. Thus, after

Table 15. Comparison of different selection threshold α on model’s performance.

α or setting	Accuracy on Each Task								Avg-Acc	Avg-Params
	SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA		
0.8	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81	0.71
0.4	78.85	59.63	96.81	55.59	56.95	24.64	62.05	59.54	61.72	0.59
1.2	79.57	59.95	96.90	56.52	57.42	28.31	65.22	61.34	63.15	0.84
every-layer	80.03	60.07	97.09	57.28	56.52	31.68	64.97	59.78	63.43	1.00

Table 16. Comparison of different feature selections on model performance.

Method	Accuracy on Each Task								Mean $\uparrow$	BWT $\uparrow$
	SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA		
Last Layer’s feature	77.55	58.17	94.50	48.91	55.45	23.40	56.40	59.44	59.23	-3.58
Pooled feature	77.67	59.19	89.03	51.06	57.03	24.69	63.43	20.68	55.34	-15.31

evaluating the excessively low performance associated with $\alpha = 0.4$ and the excessive parameter count of $\alpha = 1.2$, we selected $\alpha = 0.8$ as a more balanced value.

Impact of PTL Feature Selection. In our main results, we chose the output of the last Decoder layer for classification in the PTL. In Table 16, we also attempted to use the average pooling of outputs from all Decoder layers for classification. Classifying using features from average pooling slightly improved performance on several tasks (SQA, TQA, GQA, VizWiz, Ref), particularly achieving a 7.03% improvement on VQAv2. However, severe performance degradation ($\sim 39\%$) occurred when facing OCR-VQA. We further explored the causes of this decline and found that most samples were classified into the ImageNet task, while a small number of ImageNet samples were classified into OCR-VQA, indicating that average pooling is poor in distinguishing highly similar tasks.

Impact of LoRA Rank. We further investigated the impact of the LoRA rank on our method, as presented in the Table 17. When the LoRA rank was set to 32, there remained a certain margin for performance improvement in the model. However, when the LoRA rank reached 128, the model performance had reached a plateau, with minor performance degradation observed on specific tasks. Therefore, we selected a LoRA rank of 64 as the default setting in our method.

Impact of Model Size. We conducted experiments on models of varying scales, as delineated in Table 18. Owing to the augmentation in the parameter count of the base model, the performance of the proposed method was further enhanced. Notably, the BWT metric of our approach exhibits a higher value on the 13B model. This phenomenon may be attributed to the fact that the output of the final decoder layer in larger models encompasses more highly abstracted information (hidden-size: 5120 in the 13B model vs. 4096 in the 7B model), which facilitates PTL in achieving superior task classification performance.

Figure 5. **Confusion Matrix of PTL.** This figure illustrates the localization performance of the PTL mechanism.

Table 17. Comparison of different LoRA ranks.

Setting	Rank	Accuracy on Each Task								Mean↑	BWT↑
		SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA		
Immediate	32	78.86	59.90	96.32	55.92	57.06	24.17	63.15	60.22	61.95	–
	64	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81	–
	128	79.23	59.54	97.07	57.96	56.03	25.58	65.22	62.33	62.75	–
Last	32	77.32	58.11	94.06	48.44	55.07	22.08	54.02	57.80	58.36	-3.59
	64	77.55	58.17	94.50	48.91	55.45	23.40	56.40	59.44	59.23	-3.58
	128	77.70	57.76	94.59	50.06	54.12	23.11	56.49	59.66	59.19	-3.56

Table 18. Comparison of our method on LLaVA-v1.5-7B and LLaVA-v1.5-13B.

Setting	Model Size	Accuracy on Each Task								Mean↑	BWT↑
		SQA	TQA	ImageNet	GQA	VizWiz	Ref	VQAv2	OCR-VQA		
Immediate	7B	79.01	59.94	96.85	56.43	57.44	25.63	65.15	62.01	62.81	–
	13B	82.32	65.77	98.26	60.41	62.35	30.91	68.75	67.92	67.08	–
Last	7B	77.55	58.17	94.50	48.91	55.45	23.40	56.40	59.44	59.23	-3.58
	13B	80.79	64.82	96.01	52.93	60.86	29.11	62.18	62.27	63.62	-3.46

A.5. Generalization and Overhead Analysis

Performance on tasks that require high levels of domain-specific knowledge. We conducted an additional experiment using PMC-VQA (Zhang et al., 2023), a dataset from the medical domain which represents a significant distribution shift from the original general domain tasks. We incorporated PMC-VQA as the 9th task, trained sequentially after the original 8 tasks. The results are presented in Table 19.

It is important to acknowledge that our model’s zero-shot performance initially trails behind LLaVA-Med. This is expected, as our baseline model has not undergone large-scale pre-training specifically on biomedical corpora. However, the results demonstrate the effectiveness of our adaptation strategy: post-training, the model achieves impressive accuracy in both the Immediate evaluation and the final No-ID evaluation (facilitated by PTL), effectively bridging the performance gap.

Furthermore, this experiment provides a compelling validation of the PKGE mechanism. Due to the significant knowledge disparity between the general and medical domains, our probe successfully identified the high distribution shift and automatically triggered a substantial expansion, allocating experts to 28 out of 32 layers. This confirms that PKGE functions as intended: it conserves resources for similar tasks while aggressively expanding capacity for distinct, challenging domains to ensure optimal fit.

Table 19. Generalization capability on the medical domain (PMC-VQA).

Model / Setting	Metric (Accuracy %)	Note
LLaVA-Med	34.8	Specialized Medical Baseline
Ours (Zero-shot)	20.2	Before Adaptation
Ours (Immediate)	34.3	Adaptive Expansion (28 Experts Added)
Ours (Final)	32.1	Sequential Performance

Zero-shot evaluation. To rigorously evaluate the model’s zero-shot performance on unseen tasks, we utilized a checkpoint trained sequentially on the first six tasks (ScienceQA, TextVQA, ImageNet, GQA, VizWiz, Ref) and evaluated it directly on the seventh task (VQAv2) without any gradient updates. As shown in Table 20, the model achieved a zero-shot accuracy

of 48.32%. While this naturally trails the supervised "Immediate" performance (65.15%), it demonstrates significant generalization capability.

Crucially, a deeper inspection of the PTL mechanism reveals that approximately 80% of the VQAv2 test queries were automatically assigned to the router and experts corresponding to TextVQA. This behavior provides compelling evidence that our method operates as intended: when resolving an unknown task, the model successfully retrieves the most semantically related historical knowledge (in this case, utilizing TextVQA skills to solve VQAv2 problems) to formulate a response.

Table 20. Zero-shot performance on unseen tasks.

Evaluation Setting	Accuracy (%)	Description
Zero-shot (Ours)	48.32	Unseen task; $\sim 80\%$ routed to TextVQA experts
Immediate (Oracle)	65.15	Upper bound (evaluated immediately after training)
Sequential (Final)	56.40	Performance after completing full sequence

Evaluation of extra parameter storage and computation time overhead. Potential extra parameter storage and additional computation time introduced by using vanilla LLaVA as the feature extractor for PTL inputs. Since we already adopt LLaVA as the backbone, we can directly run inference with it and thus do not incur any extra storage cost for a separate feature extractor. For each input sample, the additional time overhead is constant-equivalent to the time required to perform one prefill pass on the input. This overhead does not grow with the length of the generated sequence and is typically within an acceptable range. We have conducted a detailed breakdown of the operations introduced by our modules. For the VAE Module, the FLOPs for a single VAE inference are approximately 100 MFLOPs, and the total FLOPs are only 0.8 GFLOPs even when eight iterative operations are considered. As for the LoRA Experts and Routers, for a single input sample with the shape of $[1, 768, 2048]$, the computational cost of each LoRA expert is about 25 MFLOPs and that of each router is about 12 MFLOPs. Even if an additional expert and a router are equipped for each of all layers (32 layers in total), the total extra computation can be calculated as $(25 + 12) \times 32 = 1184$ MFLOPs ≈ 1.184 GFLOPs. For reference, the visual encoder in LLaVA-1.5-7b needs 183.16 GFLOPs to process a single image with the resolution of 336×336 . The computation added by our method (1.184 GFLOPs) only accounts for **0.4%** of the computational cost of the visual encoder. In other words, the overhead introduced by our method is strictly lower than the marginal cost of processing a single additional image token, and is almost negligible compared with the total inference cost of the LLaVA-v1.5-7B model.

A.6. Qualitative Results

We qualitatively analyze the model’s outputs. As illustrated in Figure 6, after training on the final task, we randomly sample data from previous tasks and compare results across methods. On ImageNet, our model retains domain-specific knowledge with minimal forgetting, while MoELoRA and other methods rely on pretrained knowledge and produces generic responses. For the Ref task, our model better preserves knowledge required for non-linguistic generation.

Table 21. Comparison of extend layers on the GQA $\rightarrow$ VizWiz continual learning transition.

Method	Shared Extend Layers	Unique Extend Layers	VizWiz Score (%)
MoExtend	0, 1, 7, 9, 18, 21, 22, 23, 24, 26, 29	2, 14, 28, 30, 31	56.82
Ours	0, 1, 7, 9, 18, 21, 22, 23, 24, 26, 29	3, 12, 13, 17, 27	57.44



Figure 6. Qualitative comparison: We randomly select five datasets prone to forgetting and visualize the results of the two models.